

Project Task Introduction

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Unit Commitment Prediction in Short-term Hydropower Scheduling

Data and task provided by Sintef Energy Research

Outline

- ▶ Course project purpose
- ▶ About Sintef
- ▶ Task and dataset
- ▶ A few hints
- ▶ Kaggle scoring and ranking
- ▶ Additional computing resources
- ▶ Submission and grading guideline

Learning goals

- ▶ Practice on a realistic ML task
- ▶ ML model training lifecycle
- ▶ Experience ML competition
- ▶ Focused skills
 - ▶ Data manipulation and preparation
 - ▶ Exploratory data analysis
 - ▶ Feature engineering
 - ▶ Model ensemble
 - ▶ Model interpretation

Course project overview

- ▶ Accurately solve a prediction problem
- ▶ Real-world data
- ▶ You will compete with Virtual Teams (VTs) we've created
- ▶ Better score $\implies$ More VTs defeated $\implies$ higher grade

kaggle

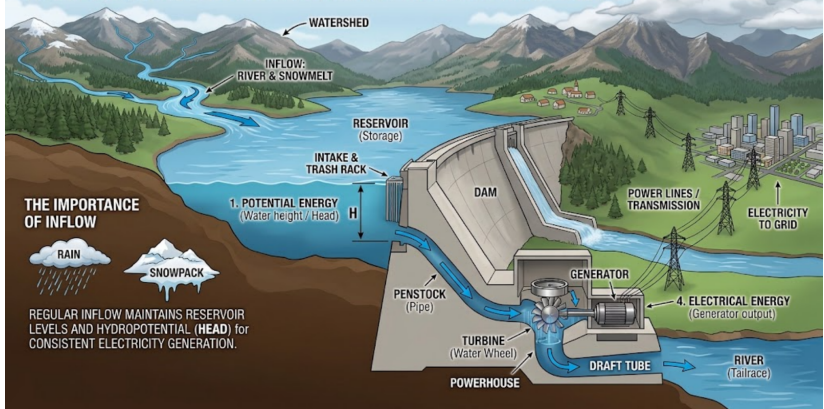
- ▶ The competition is hosted on Kaggle InClass
- ▶ Kaggle is the largest platform for data science competitions
- ▶ Why Kaggle?
 - ▶ The assessment is objective and follows the ML definition
 - ▶ It checks your solutions automatically
 - ▶ Allows private InClass competitions for students
 - ▶ User-friendly interface

SINTEF Energy Research – Strategic Focus

- Energy efficiency
- Carbon capture and storage - CCS
- Hydropower
- Offshore wind
- Bioenergy
- Transmission
- Smart grids
- Hydrogen
- Integrated energy systems
- Zero-emission transport



HOW DOES A CONVENTIONAL HYDROPOWER SYSTEM WORK?



Conventional scheduling method

- ▶ Mixed-Integer Programming (MIP)

$$\begin{aligned} \min_x \quad & c^T x \\ \text{subject to} \quad & Ax \leq b \\ & x_i \in \mathbb{Z} \text{ for some } i \\ & x_j \in \mathbb{R} \text{ for others} \end{aligned}$$

- ▶ Exact MIP: accurate but slow
- ▶ Relaxing MIP: Fast but may result in unrealistic unit commitment
 - ▶ e.g., production below minimum limits, breaking environmental constraints.
 - ▶ conventional solution: use a static heuristic (e.g., `gen_turn_off_limit = 0.5`) to avoid low production.

Unit Commitment (UC) problem (example)

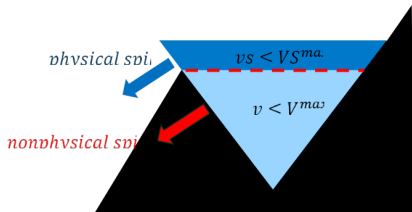
v : storage volume, vs : spill volume, V^{max} : maximum storage, VS^{max} : maximum spill amount, δ : unit commitment (0=no spill, 1=spill)

$$vs \geq 0$$

$$vs \geq v - V^{max}$$

$$vs \leq v - V^{max} \cdot \delta$$

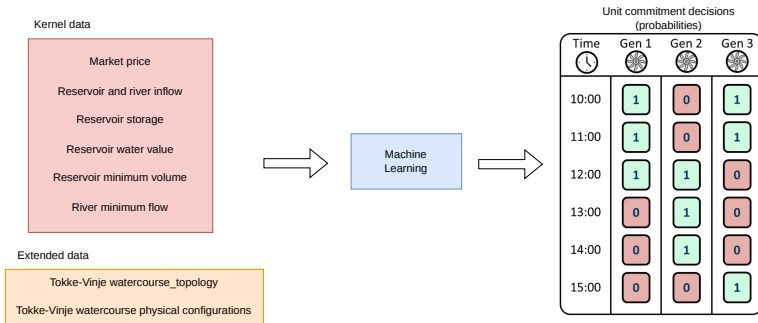
$$vs \leq VS^{max} \cdot \delta$$



- ▶ Many such binary variables in hydroscheduling
- ▶ If δ is relaxed to $[0, 1]$, the optimization becomes fast.
- ▶ But non-physical spill can happen (i.e., “spill” water before the reservoir runs full)
- ▶ ML is used to predict δ

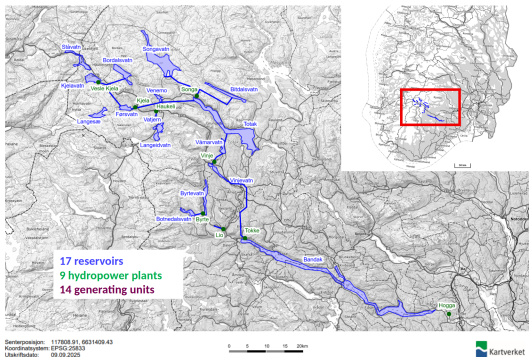
The Machine Learning problem setting

In this project, you use ML to predict whether the binary variables should be relaxed before optimization.





SINTEF open dataset: Tokke-Vinje Watercourse



- **Location:** Tokke and Vinje municipalities, Norway
- **Catchment Area:** 3,104 km²
- **Annual Inflow:** 3,697 Mm³
- **Total Annual Production:** 4.9 TWh (annual consumption of some 200,000 households)
- **Total Reservoir Volume:** 1,921 Mm³
- **Total installed capacity:** 1,003 MW
- **Total Tunnel Length:** 108 km
- **Operator:** Statkraft

A few hints related to the task

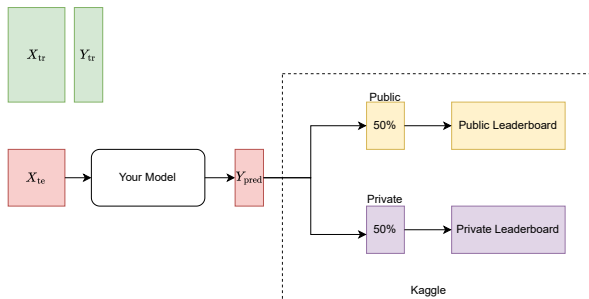
- ▶ Understand the performance evaluation metric (ROC-AUC)
- ▶ Time-aware validation
 - ▶ Random shuffling can cause information leakage in validation
- ▶ Feature engineering with temporal information
- ▶ Keep improving

General hints

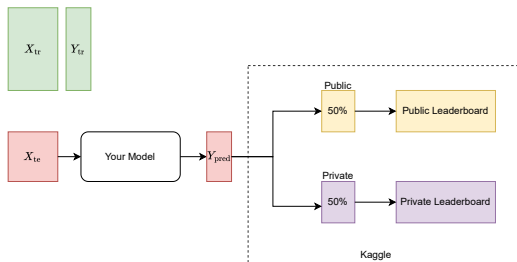
- ▶ (✓) Start early
- ▶ (✓) Work as a team; (✗) bite each other
- ▶ (✓) Generate hypotheses about the data and verify them using EDA
- ▶ (✓) Spy on other teams 😏
- ▶ (✗) Start late; submit solutions right before the deadline
- ▶ (✗) Ask for an extension
- ▶ (✗) Write messy notebooks and dirty code
- ▶ (✗) Scream in despair
- ▶ (✗) Cheat and use external data

Kaggle Scoring and Ranking

- ▶ You use X_{tr} , Y_{tr} , and maybe X_{te} to create your model
- ▶ You apply your model on X_{te} to get the predictions Y_{pred} and submit them to Kaggle
- ▶ Kaggle compares your predictions to the ground truths in the public and private halves accordingly.
- ▶ You see the public score and ranking immediately
- ▶ You see the private score and ranking after the deadline



Important Remarks



- ▶ Your course base points are calculated with the **private leaderboard**.
- ▶ A good public ranking does not necessarily lead to a good private ranking
 - ▶ Public scores are only estimates
- ▶ The data splits
 - ▶ Training: 2015-01-01 to 2023-01-01
 - ▶ Testing (public): 2023-01-01 to 2024-01-01
 - ▶ Testing (private): 2024-01-01 to 2024-12-25

Additional Computing Resources

- ▶ Computing resources are important in ML
- ▶ IDI provides some help
 - ▶ Every project group has a virtual machine
`tdt4173-xxx.idi.ntnu.no`
 - ▶ xxx is your assigned machine number
 - ▶ 4 CPUs + 32GB RAM (no GPU)
 - ▶ You can install software there
 - ▶ Will be ready after the groups are formed
- ▶ Free cloud credits
 - ▶ face value \$100 for Microsoft and \$50 for Google
 - ▶ We provide tutorial PDFs and videos in Canvas
- ▶ Contact Amalie S. Johannessen
(amalie.s.johannessen@ntnu.no) and Thomas Heggø Teigland
(thomas.teigland@ntnu.no) for help

Submission and Grading Guideline

Read carefully the PDF in Canvas